Behavioral Customer Segmentation Analysis

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Introduction

Understanding customer behavior is a critical challenge for organizations seeking to improve targeting, personalization, and customer engagement. This project builds an end-to-end customer segmentation system that identifies distinct behavioral customer groups using unsupervised machine learning. By combining demographic information, purchasing activity, campaign responsiveness, and channel engagement data, the solution transforms raw customer records into actionable customer personas. Beyond clustering, the system operationalizes segmentation through business recommendations, workflow suggestions, and API-based deployment, enabling customer intelligence to support real business decision-making.

Objectives

What am I trying to find out?

- Can customer behavioral patterns be used to identify meaningful customer segments?
- Which groups of customers exhibit similar spending, engagement, and purchasing behaviors?
- How can customer segments be translated into actionable business intelligence and marketing strategies?

What do I already know?

- Customers do not behave uniformly across purchasing channels and campaigns.
- Behavioral differences often exist even among customers with similar demographic characteristics.
- Customer segmentation becomes more valuable when connected to business actions rather than cluster labels alone.

What am I aiming to achieve?

- Build an end-to-end customer segmentation system using unsupervised machine learning.
- Discover meaningful behavioral personas through clustering analysis.
- Translate technical cluster outputs into business-friendly customer segments.
- Generate recommendations and workflow suggestions that support marketing and CRM activities.
- Deploy the solution through an API that enables scalable customer intelligence generation.

What factors influence the results?

- The dataset represents customer purchasing, engagement, and demographic behavior at a specific point in time.
- Segment quality depends heavily on feature engineering and behavioral signal creation.
- Clustering results are sensitive to feature scaling, transformation choices, and cluster selection methodology.
- Business usefulness depends on how effectively clusters can be interpreted and converted into actionable personas.

Data Preparation

The dataset was obtained from Kaggle and contains approximately 2,240 customer records covering demographic characteristics, purchasing behavior, campaign responses, and channel engagement activity. During preparation, non-informative columns (ID, Z_Revenue, and Z_CostContact) were removed to eliminate irrelevant information from the analysis. The customer enrollment date (Dt_Customer) was transformed into a more meaningful behavioral feature, tenure_years, representing the customer's relationship length relative to the most recent customer in the dataset, after which the original date column was removed. Categorical variables (Education and Marital_Status) were explicitly assigned string data types to ensure consistent downstream processing. The resulting dataset consisted of 26 features spanning customer demographics, spending activity, purchasing channels, campaign interactions, household composition, and customer tenure, providing a structured foundation for exploratory analysis and segmentation modeling.

Exploratory Data Analysis (EDA)

The exploratory analysis focused on evaluating the dataset's suitability for behavioral segmentation and identifying factors that could influence cluster formation. Distribution visualizations and skewness analysis revealed heavily right-skewed spending and purchasing variables, while scale and distance assessments highlighted substantial differences in feature magnitudes that would later require normalization. Spearman correlation analysis was used to examine relationships between variables due to the non-normal nature of many features.

Additional investigation included zero-inflation analysis for sparse behavioral indicators, categorical feature auditing covering cardinality, frequency distributions, and rarecategory detection, and outlier visualization using distribution-based techniques. Although the IQR method identified a significant number of outliers, these observations were retained because they likely represented legitimate customer behaviors such as high-value or highly engaged customers rather than data quality issues.

Missing values were limited to the Income feature and were addressed through median imputation, providing a robust solution while preserving the overall distribution of customer income. Together, these analyses established a reliable foundation for feature engineering, transformation, and clustering model development.

Feature Engineering

Feature engineering was performed to transform raw customer attributes into higher-level behavioral indicators that better represent purchasing habits, engagement patterns, channel preferences, household characteristics, and customer loyalty. The objective was to create features that capture meaningful customer behaviors not directly observable from the original dataset while improving the interpretability and business relevance of the resulting customer segments.

Feature Name	Definition / Logic	What It Captures	Notes / Assumptions
total_spending	Sum of all spending categories	Overall customer monetary value	Foundational aggregate used by multiple derived features
total_purchases	Sum of web and store purchases	Purchase frequency and activity level	Enables ratios and channel-behavior indicators
is_web_dominant	NumWebPurchases / total_purchases > 0.5	Primary purchase channel preference	Binary dominance simplifies interpretation
value_optimizer	High total spending and high deal-purchase ratio	High spenders who actively seek discounts	Composite behavioral feature combining value and promotion sensitivity

Feature Name	Definition / Logic	What It Captures	Notes / Assumptions
spending_to_income_ratio	total_spending / Income	Spending intensity relative to purchasing power	Normalizes spending behavior by customer income
family_load	Kidhome + Teenhome grouped into bins	Household responsibility level	Coarse categories improve interpretability and reduce noise
is_long_tenure	tenure_years ≥ dataset median	Relationship maturity with brand	Tenure treated as a relative measure within the dataset rather than an absolute threshold

Feature Selection

Feature selection was performed using an unsupervised Leave-One-Feature-Out (LOFO) methodology designed specifically for clustering problems. Because no target variable exists in unsupervised learning, feature importance was evaluated based on each variable's contribution to cluster quality rather than predictive performance. A baseline K-Means model was first evaluated using the Silhouette Score, after which each feature was removed individually and the clustering process repeated. The resulting change in silhouette score served as a proxy measure of feature importance, indicating how much each variable contributed to cluster separation and cohesion. This analysis revealed that customer tenure, financial capacity, purchasing behavior, and promotional engagement were among the strongest drivers of cluster structure, while several features provided limited additional information or introduced redundancy. Based on these findings, low-contributing variables were removed to simplify the feature space, reduce noise, and improve the interpretability and stability of the final customer segmentation model.

Transformation & Scaling

Feature transformation and scaling were performed to improve the geometric suitability of the dataset for distance-based clustering. Skewness analysis identified several heavily right-skewed financial, spending, and purchasing variables that could disproportionately influence cluster formation. To address this issue, logarithmic (log1p) transformations were applied to selected features, reducing the impact of extreme values while preserving the relative ordering of observations. The effectiveness of these transformations was validated through a before-and-after skewness assessment, which demonstrated substantially more balanced distributions across the transformed variables. Following transformation and feature selection, all variables were standardized using StandardScaler to place them on a common scale, ensuring that no single feature dominated distance calculations due to differences in magnitude. This process produced a balanced and normalized feature space suitable for robust K-Means clustering and customer segmentation.

Model Training & Cluster Selection

The segmentation model was developed using K-Means clustering and evaluated across multiple candidate cluster counts. The Elbow Method was first used to identify a reasonable range of cluster solutions by analyzing the reduction in within-cluster variance as additional clusters were introduced. Candidate models were then assessed using Silhouette Analysis to evaluate cluster cohesion and separation. While the highest silhouette score was achieved at K = 2, indicating the strongest natural mathematical split in the dataset, the resulting segmentation produced only broad customer groupings with limited business value. Additional candidate solutions (K = 4, K = 5, and K = 6) were profiled and compared based on cluster quality, size distribution, behavioral distinctiveness, and business interpretability. The K = 6 solution was ultimately selected as the champion model because it provided the richest behavioral segmentation while maintaining stable cluster sizes and meaningful business differentiation. Following model selection, each cluster was profiled and translated into an operational customer persona with associated business value and recommended actions.

Cluster	Segment Name	Behavioral Traits	Business Value	Recommended Action
0	Dormant Low-Value Customers	Low spending, weak purchasing activity, low engagement across channels, minimal campaign responsiveness	Low immediate revenue contribution but potential reactivation opportunity	Launch re-engagement campaigns, simplified offers, and low-cost retention workflows
1	Active Omni-Channel Customers	Strong purchasing activity across web, catalog, and store	Stable and valuable customer base	Maintain engagement through personalized recommendations and cross-channel marketing

Cluster	Segment Name	Behavioral Traits	Business Value	Recommended Action
		channels with balanced engagement	with consistent activity	
2	Premium Loyal Customers	High spending behavior, strong catalog purchases, high engagement, premium behavioral profile	High customer lifetime value and strong retention potential	Prioritize loyalty programs, exclusive offers, and premium customer experiences
3	Family Budget Customers	Family-oriented purchasing patterns with lower overall spending and moderate engagement	Moderate long-term value with price-sensitive behavior	Promote bundles, family-oriented offers, and value-driven campaigns
4	Deal-Driven Customers	High promotion and discount sensitivity, active deal purchasing behavior	Strong transactional activity but potentially lower margins	Use targeted discounts, flash sales, and promotion-based automations strategically
5	VIP Campaign Responders	Extremely high campaign responsiveness combined with strong spending behavior	Highly marketable and highly responsive premium segment	Trigger VIP workflows, early-access campaigns, premium retention strategies, and high-priority personalization